Dr. Yonatan Ashenafi

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Education

Rensselaer Polytechnic Institute (RPI), Troy, NY
Ph.D. in Mathematics, May 2021
Thesis: Stochastic Hydrodynamics of Colonial Microswimmers

M.S. in Applied Mathematics, August 2016 – December 2018

Dordt University, Sioux Center, IA
B.A. in Mathematics, August 2012 – May 2016

Experience

Postdoctoral Fellow, Worcester Polytechnic Institute (WPI) *Aug 2023 – Jul 2025*
Worcester, MA

- Conducting research on modeling intracellular transport processes using experimental data from multi-university collaborations.
- Teaching Multivariable Calculus, Calculus I & II, and Ordinary Differential Equations.

Postdoctoral Fellow, University of Alberta *Sep 2021 – May 2023*
Edmonton, AB

- Researched motion mechanisms in diatoms (microscopic algae of ecological importance) through video-based data analysis.
- Taught Calculus I for Life Sciences and Calculus II for Physical Sciences.

Technical Fellow Intern, General Electric (GE) Research *Oct – Dec 2020*
GE Global Research Center, Niskayuna, NY

- Collaborated with the Probabilistic Design Team on experimental design and additive manufacturing projects.
- Developed Python programs for reinforcement learning tasks.

Research Assistant, Dordt College *Jun – Aug 2014*
Sioux Center, IA

- Constructed Gaussian mixture models from bacterial gene expression data using R.
- Gained experience in statistical data analysis and biological data modeling.

Tutor, Dordt College Library *Sep 2013 – May 2016*
Sioux Center, IA

- Tutored students in College Algebra, Calculus I–III, Linear Algebra, and Discrete Mathematics.
- Collaborated effectively with students, faculty, and staff to improve academic performance.

Service and Activities

- Proposal Reviewer, National Science Foundation (NSF), 2025–present.
- Presenter, 2025 Canadian Mathematical Society Mathematics Education Meeting (Online)
- Judge, SIMIODE SCUDEM X 2025 (Differential Equations Modeling Challenge).
- Volunteer Grader, WPI Annual Math Meet, 2024.
- Judge, University of Alberta Undergraduate Research Symposium, 2022.
- Presenter, Ethiopian Mathematics Professional Association Seminar, July 2025.
- Treasurer, Black Graduate Students Association, RPI (2020). Managed finances during the COVID-19 pandemic.
- Organizer, Mini-Symposium on Noise and Asymmetry in Microscopic Life, CAIMS 2022.
- Mathematical Modeler, Mathematical Problems in Industry (MPI) Workshop, Univ. of Vermont (2024) and RPI (2017).
- Participant, SIMIODE Spring 2024 Webinars.
- Presenter at major conferences: Biophysical Society (2025), BAMM (2024), SIAM Life Sciences (2021), MBI Workshop (2020), and Univ. of Minnesota Multiscale Modeling (2019).

Awards

- Bill and Nancy Siegmann Applied Mathematical Modeling Prize, RPI (2021).
- DiPrima Summer Research Fellowship, RPI (2018).
- ASME CIE Advanced Modeling and Simulation Best Paper Award (2022).

Publications

- Ashenafi, Y. & Kramer, P. R. (2024). *Statistical Mobility of Multicellular Colonies of Flagellated Swimming Cells.* **Bulletin of Mathematical Biology**.
- Ashenafi, Y. (2022). *Stability and Spatial Autocorrelations of Suspensions of Microswimmers with Heterogeneous Spin.* **Physics of Fluids**.
- Ashenafi, Y., Pandita, P., & Ghosh, S. (2021). *Reinforcement Learning-Based Sequential Batch-Sampling for Bayesian Optimal Experimental Design.* **Journal of Mechanical Design**.
- Zheng, P. et al. (2023). *Cooperative Motility, Force Generation and Mechanosensing in a Foraging Non-Photosynthetic Diatom.* **Open Biology**, 13(10), 230148.
- Disselkoen, C., et al. (2016). *A Bayesian Framework for the Classification of Microbial Gene Activity States.* **Frontiers in Microbiology**, 7, 1191.
- Ashenafi, Y. & Kramer, P. R. *Asymptotic Analysis of Kinesis and Taxis for Colonial Protozoa.* Under review.
- Ashenafi, Y. & Newby, J. *Adhesion and Geometry Govern Mobility in Raphid Diatoms.* In preparation.

- Ashenafi, Y. & Walcott, S. *Simulation-Based Inference of Hidden Motor States in Intracellular Cargo Transport*. In preparation.

Technical Skills

Python (NumPy, Pandas, SciPy, TensorFlow, scikit-learn), MATLAB, C++, R, SQL

Professional Membership

Society for Industrial and Applied Mathematics (SIAM)